

Skills Summary

Does the Data Really Show a Cause?

Use this reference while completing your worksheet or when studying.

Skill 1 — Computing the statistic that was asked for

Mean: add the values and divide by how many there are.

Percentage of a group: $\frac{\text{count in the group}}{\text{size of that group}} \times 100$.

The denominator is the group the question is about, not the whole sample. Choosing the whole sample makes every group look small and is the commonest arithmetic error on these questions.

Example: Of 120 people surveyed, 45 own a bicycle, and 36 of those cyclists exercise daily. What percent of the cyclists exercise daily?

Step 1. The question is about the cyclists, so the group size 45 is the denominator, not 120.

Step 2. $\frac{36}{45} \times 100 = 0.8 \times 100 \Rightarrow$ **80**

Try it: Of 90 people surveyed, 28 own a dog, and 21 of those owners walk daily. What percent of the dog owners walk daily?

check: 25

Skill 2 — Association is not a cause

Association: the two quantities move together in the data — a pattern, and only that.

Cause: changing one would change the other. Data gathered by *watching* can never show this on its own. Three rival explanations always remain: a **third factor** moving both, the causation running **backwards**, or plain **coincidence**.

Example: Three towns report 3, 6 and 9 cases. How much bigger is the largest than the smallest, and what does that gap show?

Step 1. The gap is a subtraction; deciding what it means is a separate question the numbers cannot answer.

Step 2. $9 - 3$ gives the gap. It shows the towns differ, and nothing at all about what made them differ. $\Rightarrow$ **6**, a real difference but not a cause

Try it: Three towns report 5, 9 and 14 cases. How much bigger is the largest than the smallest?

check: 6

Skill 3 — What evidence would settle it

A fair experiment: the researcher decides *at random* who goes into each group, so the groups start out alike and the treatment is the only planned difference. Then a gap between the group means can be attributed to the treatment.

In a survey, people sort themselves, so whatever made them choose could also explain the gap.

Example: In a randomised trial the treated group's mean was 64 and the comparison group's mean was 58. How large is the gap, and may it be called an effect?

Step 1. Comparing two groups means subtracting their means; whether the gap counts as an effect depends on how the groups were formed.

Step 2. $64 - 58$ gives the gap, and because the groups were assigned at random the treatment is the only planned difference between them. $\Rightarrow$ **6** marks, which here may fairly be called an effect

Try it: In a randomised trial the treated group's mean was 71 and the comparison group's mean was 63. How large is the gap?

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(!) Watch out: “the numbers go up together” is the beginning of an argument, never the end of one. Before accepting a cause, ask what else could move both quantities, and whether anyone actually changed anything.